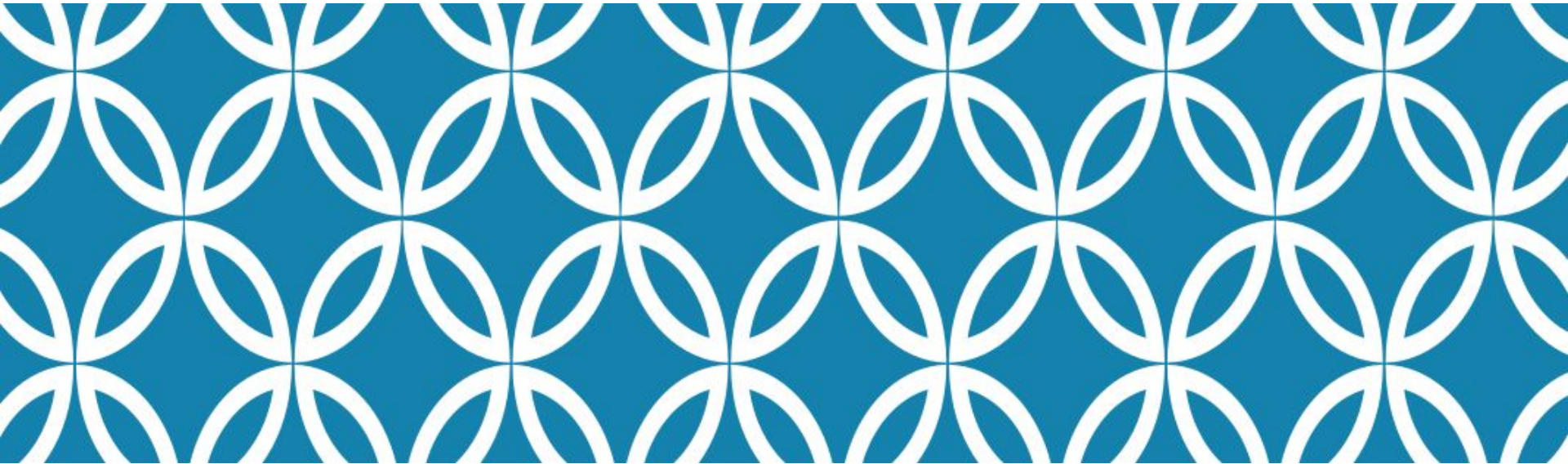


Database / Base de Données



LESSON 2 Relational Algebra

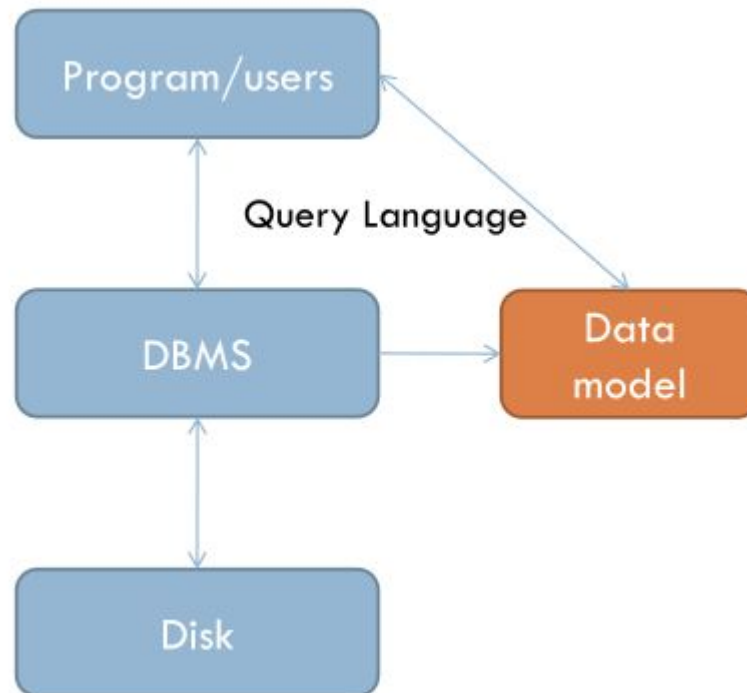
Prepared by Mr. Nop Phearum
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Planning

- ❏ Introduction
- ❏ Example database
- ❏ Basic operation
- ❏ Additional operation
- ❏ Complex query

Introduction

DBMS stores data and allows data manipulation.



Introduction

- ❏ Query language (QL): language used to retrieve and manipulate data in database.
- ❏ Query language supported by RDBMS:
 - ❏ Relational algebra
 - ❏ Relational calculus

Example database

Teacher

<u>TID</u>	Tname	DoB	Degree	Field	<u>Dnam</u>
1	Sok Dara	01/01/85	Master	Math	Math
2	Sam Sambath	01/02/80	PhD	Mechanic of fluid	Mechanic
3	Sao Piseth	05/08/70	PhD	Biology	Environment
4	Tao Pisey	14/07/65	Engineer	Electronic	Electronic
5	Van Dany	08/12/87	Engineer	hydropower	Environment

Department

<u>Dname</u>	Description
Math	...
Electronic	...
Mechanic	...
Environment	...

Children

<u>TID</u>	<u>ChildName</u>	DoB
1	Dan	01/01/2010
1	Paul	01/02/2011
2	Heng	09/08/2000
3	Lundi	08/08/1995
4	Dimanche	07/12/1987

StudentCourse

<u>StudentID</u>	<u>CID</u>
1	C1
1	C2
2	C2
2	C1
3	C2

Course

<u>CID</u>	Cname
C1	Math
C2	Physic

Student

<u>StudentID</u>	Sname
1	Chea Nary
2	Sok Palin
3	Yi Sophea

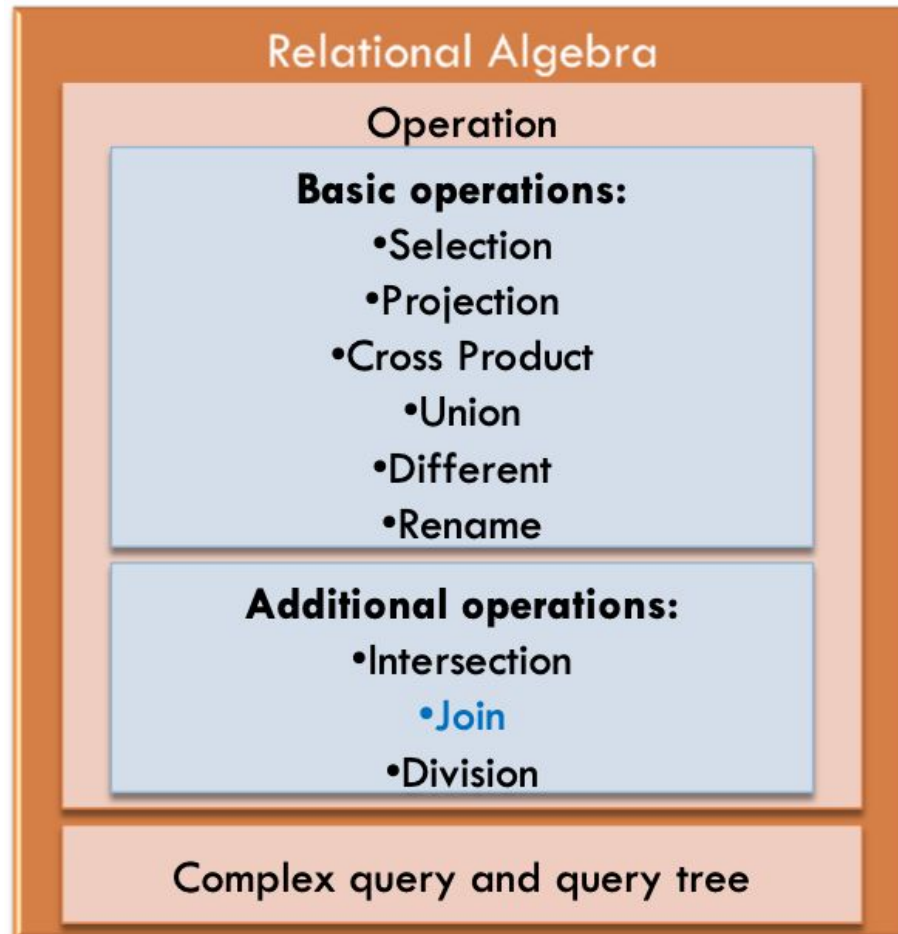
Example database

- ❑ Find out teachers who get PhD degree.
- ❑ Find out teacher who teach in department which has the same name as his/her skill.
- ❑ Show name and skill of each teacher.
- ❑ What if every student studies all course.
- ❑ Show all teacher in the city.
- ❑ Show teachers who work for ITC, don't work for RUPP.
- ❑ Show teachers who work for both ITC and RUPP.
- ❑ Show all teacher and their children.
- ❑ Find out teachers whom younger than a child of other teacher.
- ❑ Find out student who study all course.

How can we get the answer?

Relational Algebra

Operation based query language.



Relational Algebra

- ❑ Take one or two relation as argument and return a relation as output
- ❑ Operation based query language
- ❑ Complex query is formed by combining basic operations
- ❑ Each query describe procedure for computing answer

Basic Operation

- ❑ Selection($\sigma_a\theta_b(R)$ or $\sigma_a\theta_v(R)$): select a subset of rows from a relation; retrieve certain rows
 - ❑ R: a relation; a, b: attribute of R
 - ❑ θ : comparison operation ($=, \neq$ or $<>, >, <, \geq, \leq$)

$\sigma_{Degree="PhD"}(Teacher)$

<u>TID</u>	Tname	DoB	Degree	Field	<u>Dname</u>
2	Sam Sambath	01/02/80	PhD	Mechanic of fluid	Mechanic
3	Sao Piseth	05/08/70	PhD	Biology	Environment

$\sigma_{Field=Dname}(Teacher)$

<u>TID</u>	Tname	DoB	Degree	Field	<u>Dname</u>
1	Sok Dara	01/01/85	Master	Math	Math
4	Tao Pisey	14/07/65	Engineer	Electronic	Electronic

Basic Operation

- ❏ Select
 - ❏ Find out teacher who was born after 1985

Teacher

<u>TID</u>	Tname	DoB	Degree	Field	<u>Dnam</u>
1	Sok Dara	01/01/85	Master	Math	Math
2	Sam Sambath	01/02/80	PhD	Mechanic of fluid	Mechanic
3	Sao Piseth	05/08/70	PhD	Biology	Environment
4	Tao Pisey	14/07/65	Engineer	Electronic	Electronic
5	Van Dany	08/12/87	Engineer	hydropower	Environment

Basic Operation

❏ Projection ($\pi_{a,\beta,\dots}(R)$)

- ❏ return only the wanted attributes and their values; retrieve certain columns

$\pi_{Tname, Field}(Teacher)$

Tname	Field
Sok Dara	Math
Sam Sambath	Mechanic of fluid
Sao Piseth	Biology
Tao Pisey	Electronic
Van Dany	hydropower

Basic Operation

- ❑ Projection ($\pi_{a,\beta,\dots}(R)$)
 - ❑ List out degree of all teacher

$\pi_{Degree}(Teacher)$

Degree
Master
PhD
Engineer

Basic Operation

Teacher

<u>TID</u>	Tname	DoB	Degree	Field	<u>Dnam</u>
1	Sok Dara	01/01/85	Master	Math	Math
2	Sam Sambath	01/02/80	PhD	Mechanic of fluid	Mechanic
3	Sao Piseth	05/08/70	PhD	Biology	Environment
4	Tao Pisey	14/07/65	Engineer	Electronic	Electronic
5	Van Dany	08/12/87	Engineer	hydropower	Environment

Write relational algebra (RA) for following the question.

❑ Which department does Sok Dara work in?



Basic Operation

- ❑ Cross Product (X): each row of R is paired with each row of S

❑ $R(a_1, a_2, a_3); S(b_1, b_2, b_3)$

$R \times B = T(a_1, a_2, a_3, b_1, b_2, b_3)$

StudentCourse X Course

StudentCourse=SC			Course=C		10				
5	<u>StudentID</u>	<u>CID</u>	2	<u>CID</u>	<u>Cname</u>	2	C2	C1	Math
	1	C1		2	C2	C2	Physic		
	1	C2	2	C1	C1	Math			
	2	C2	2	C1	C2	Physic			
	2	C1	3	C2	C1	Math			
3	C2				3	C2	C2	Physic	

Basic Operation

- ❑ Cross Product (X)
 - ❑ List all student's ID and name of course they enrolled

$$\sigma_{SC.CID=C.CID}(SC \times C)$$

<u>SC.StudentID</u>	C.Cname
1	Math
1	Physic
2	Physic
2	Math
3	Physic

<u>SC.StudentID</u>	<u>SC.CID</u>	<u>C.CID</u>	C.Cname
1	C1	C1	Math
1	C2	C2	Physic
2	C2	C2	Physic
2	C1	C1	Math
3	C2	C2	Physic

$$\pi_{StudentID, Cname}(\sigma_{SC.CID=C.CID}(SC \times C))$$

Basic Operation

❏ Cross Product (X)

- ❏ List all student's name and the name of course that they enrolled in

Course

<u>CID</u>	Cname
C1	Math
C2	Physic

Student

<u>StudentID</u>	Sname
1	Chea Nary
2	Sok Polin
3	Yi Sophea

StudentCourse

<u>StudentID</u>	<u>CID</u>
1	C1
1	C2
2	C2
2	C1
3	C2



Basic Operation

- ❏ Union ($R \cup S$) : return all rows occur in R or S
 - ❏ R and S must be union compatible:
- ❏ Schema of R and S have the same degree
 - ❏ Attributes in the same order must have the same data type (domain)
- ❏ Set Different ($R - S$): Return rows occur in R but not in S. R and S must be union compatible.

Basic Operation

Teacher

<u>TID</u>	Tname	DoB	Degree	Field	<u>Dnam</u>
1	Sok Dara	01/01/85	Master	Math	Math
2	Sam Sambath	01/02/80	PhD	Mechanic of fluid	Mechanic
3	Sao Piseth	05/08/70	PhD	Biology	Environment
4	Tao Pisey	14/07/65	Engineer	Electronic	Electronic
5	Van Dany	08/12/87	Engineer	hydropower	Environment

Teacher1

<u>TID</u>	Tname	DoB	Degree	Field	<u>Dnam</u>
1	Sok Dara	01/01/85	Master	Math	Math
12	Nicole John	01/02/87	Master	BD	CS
3	Sao Piseth	05/08/70	PhD	Biology	Environment
14	Michael Jam	14/07/81	Engineer	Electronic	Electronic

Basic Operation

Teacher $\vee$ Teacher

<u>1ID</u>	Tname	DoB	Degree	Field	<u>Dnam</u>
1	Sok Dara	01/01/85	Master	Math	Math
2	Sam Sambath	01/02/80	PhD	Mechanic of fluid	Mechanic
3	Sao Piseth	05/08/70	PhD	Biology	Environment
4	Tao Pisey	14/07/65	Engineer	Electronic	Electronic
5	Van Dany	08/12/87	Engineer	hydropower	Environment
12	Nicole John	01/02/87	Master	BD	CS
14	Michael Jam	14/07/81	Engineer	Electronic	Electronic

Disjunction

Teacher - Teacher 1

<u>TID</u>	Tname	DoB	Degree	Field	<u>Dnam</u>
2	Sam Sambath	01/02/80	PhD	Mechanic of fluid	Mechanic
4	Tao Pisey	14/07/65	Engineer	Electronic	Electronic
5	Van Dany	08/12/87	Engineer	hydropower	Environment

Negation

Additional operation

- ❏ Intersection ($R \cap S$): Return all rows that occur in both R and S. R and S must be Union compatible.
- ❏ Teacher $\cap$ Teacher 1

<u>TID</u>	Tname	DoB	Degree	Field	<u>Dnam</u>
1	Sok Dara	01/01/85	Master	Math	Math
3	Sao Piseth	05/08/70	PhD	Biology	Environment

Additional operation

- Put the right operation for the result

R1

Name	Age	Sex
A	20	M
C	21	M
B	21	F

R2

Name	Age	Sex
D	20	F
A	20	M
E	21	F

R1 R2 =

Name	Age	Sex
A	20	M
C	21	M
B	21	F
D	20	F
E	21	F

R1 R2 =

Name	Age	Sex
A	20	M

R1 R2 =

Name	Age	Sex
C	21	M
B	21	F

Additional operation

- ❑ Join: used to combine information from many tables.
 - ❑ can be expressed using Cross Product, selection and projection
- ❑ Condition join or Theta join ($R \bowtie_c S$) = $(\sigma_c(R \times S))$: selection of cross product based on condition c.
 - ❑ $R(a_1, a_2, a_3); S(b_1, b_2, b_3)$
 - ❑ $R \bowtie_c S = T(a_1, a_2, a_3, b_1, b_2, b_3)$

$$\begin{aligned} & Teacher \bowtie_{Teacher.TID=Children.TID} Children \\ &= \sigma_{Teacher.TID=Children.TID}(Teacher \times Children) \end{aligned}$$

Additional operation

<u>TID</u>	Tname	DoB	Degree	Field	<u>Dnam</u>
1	Sok Dara	01/01/1985	Master	Math	Math
2	Sam Sambath	01/02/1980	PhD	Mechanic of fluid	Mechanic
3	Sao Piseth	05/08/1970	PhD	Biology	Environment
4	Tao Pisey	14/07/1965	Engineer	Electronic	Electronic
5	Van Dany	08/12/1987	Engineer	hydropower	Environment

<u>TID</u>	<u>ChildName</u>	DoB
1	Dan	01/01/2010
1	Paul	01/02/2011
2	Heng	09/08/2000
3	Lundi	08/08/1995
4	Dimanche	07/12/1987

Teacher Teacher ⋈_{Teacher.DoB > Children.DoB} Children

= $\sigma_{\text{Teacher.DoB} > \text{Children.DoB}} (\text{Teacher} \times \text{Children})$

<u>T.TID</u>	T.Tname	T.DoB	T.Degree	T.Field	<u>T.Dnam</u>	<u>C.TID</u>	C.ChildName	C.DoB
5	Van Dany	08/12/87	Engineer	hydropower	Environment	4	Dimanche	07/12/1987

Additional operation

❑ Equi-Join: special case of condition join where only equalities are allowed in condition.

❑ $R(a1, x, a3); S(b1, b2, y)$

❑ $R \bowtie_{x=y} S = T(a1, x, a3, b1, b2)$

⋈ Teacher.DoB > Children.DoB Children

<u>T.TID</u>	T.Tname	T.DoB	T.Degree	T.Field	<u>T.Dnam</u>	C.ChildName	C.DoB
1	Sok Dara	01/01/85	Master	Math	Math	Dan	01/01/2010
1	Sok Dara	01/01/85	Master	Math	Math	Paul	01/02/2011
2	Sam Sambath	01/02/80	PhD	Mechanic of fluid	Mechanic	Heng	09/08/2000
3	Sao Piseth	05/08/70	PhD	Biology	Environment	Lundi	08/08/1995
4	Tao Pisey	14/07/65	Engineer	Electronic	Electronic	Dimanche	07/12/1987

Additional operation

❑ Natural join ($R \bowtie S$): equi-join on all common attributes (attributes with the same name).

❑ $R(a1, a2, x, y); S(b1, x, y, b2, b3)$

$R \bowtie S = T(a1, a2, x, y, b1, b2, b3)$

❑ x, y : are common attributes of R and S

$StudentCourse \bowtie Course$

<u>StudentID</u>	<u>CID</u>
1	C1
1	C2
2	C2
2	C1
3	C2

$\bowtie$

<u>CID</u>	Cname
C1	Math
C2	Physic

<u>StudentID</u>	<u>CID</u>	Cname
1	C1	Math
1	C2	Physic
2	C2	Physic
2	C1	Math
3	C2	Physic

Additional operation

Join

- List all students' name and the name of the course they enroll in.

Course

<u>CID</u>	Cname
C1	Math
C2	Physic

Student

<u>StudentID</u>	Sname
1	Chea Nary
2	Sok Polin
3	Yi Sophea

StudentCourse

<u>StudentID</u>	<u>CID</u>
1	C1
1	C2
2	C2
2	C1
3	C2



Additional operation

- ❑ Division (R/S or $R \div S$): used to express certain kind of question like:
“List students who enroll to all courses”.
- ❑ Can be expressed using projection, cross product, and different.
- ❑ $R(x, y); S(y)$
$$R \div S = \pi_x(R) - \pi_x((\pi_x(R) \times S) - R)$$
- ❑ $\text{StudentCourse} \div \text{Course}$

What does the query really mean?

- ❑ $(\text{studentID}, \text{CID}) \div \text{CID} = \text{studentID}$

Additional operation

StudentCourse

<u>StudentID</u>	<u>CID</u>
1	C1
1	C2
2	C2
2	C1
3	C2

Course

<u>CID</u>	Cname
C1	Math
C2	Physic

$\pi_{studentID}(StudentCourse) \times \pi_{CID}(Course)$

<u>StudentID</u>	<u>CID</u>
1	C1
1	C2
2	C1
2	C2
3	C1
3	C2

StudentCourse $\div$ Course

$= \pi_{studentID}(StudentCourse)$

- $\pi_{studentID}((\pi_{studentID}(StudentCourse) \times \pi_{CID}(Course)))$
- StudentCourse

<u>StudentID</u>
1
2

Additional operation

- ❑ Rename(): used to rename attribute and relation.
- ❑ Solve the problem of attribute conflict in cross product and join operation.
- ❑ All self-join and self cross product
- ❑ Rename attribute R.a to X, S.a to y, and the result of RXS to T

$$\rho(T(R.a \rightarrow x, s.a \rightarrow y), R \times S)$$

Additional operation

$\rho(T(Teacher.DoB \rightarrow TDoB, Children.DoB$
 $\rightarrow CDoB), Teacher \bowtie_{Teacher.TID=Children.TID} Children)$

<u>TID</u>	Tname	TDoB	Degree	Field	<u>Dnam</u>	ChildName	CDoB
1	Sok Dara	01/01/85	Master	Math	Math	Dan	01/01/2010
1	Sok Dara	01/01/85	Master	Math	Math	Paul	01/02/2011
2	Sam Sambath	01/02/80	PhD	Mechanic of fluid	Mechanic	Heng	09/08/2000
3	Sao Piseth	05/08/70	PhD	Biology	Environment	Lundi	08/08/1995
4	Tao Pisey	14/07/65	Engineer	Electronic	Electronic	Dimanche	07/12/1987

Complex query

- ❑ Complex query is composed of many operations
 - ❑ Ex: List the name of teacher who was not a student of the school

$$\pi_{Tname}(\pi_{Tname, DoB}(Teacher) - \pi_{Tname, DoB}(Student))$$

Reference

- ❏ R. Ramakrishnan et J. Gehrke Database Management Systems. 2003
(troisième édition)
- ❏ Jean-Marc Petit(2011). Fondamentaux de la modélisation des données.
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